GRADE

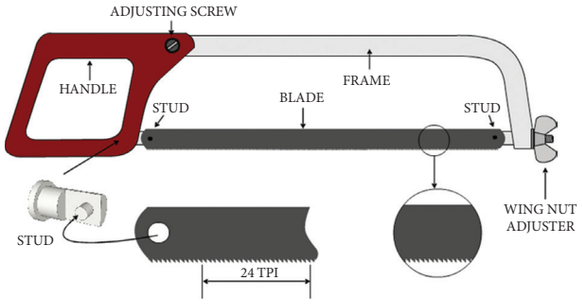
The grade of a file refers to the coarseness or the spacing between the rows of the teeth. It is designated by the number of rows of teeth per inch.

These are five types of files according to its grade. They are,

1. Rough file (R)-20-25 teeth/inch
2. Bastard file (B)-25-30 teeth/inch
3. Second cut file (SC)-35-40 teeth/inch
4. Smooth file (S)-40-60 teeth/inch
5. Dead smooth file (DS)-80-100 teeth/inch.

2.6 HACK SAW FRAME

Hack saw frame consist of a frame, a wooden handle, prongs, tightening screw and a wing nut. It is used for sawing all metals expect hardened steel. Tightening screw with the help of a wing nut is used to stretch the blade is desired.



There are two types of hacksaw namely,

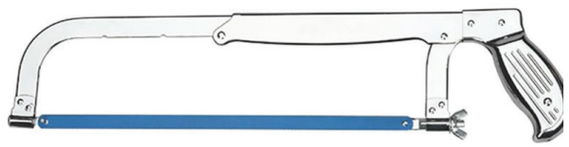
1. Standard or solid hacksaw
2. Adjustable hacksaw

STANDARD HACKSAW

In this type, the distance between the prongs cannot be altered. So it is suitable for a particular length of hacksaw blades only.

ADJUSTABLE HACKSAW

In this type, the distance between the prongs can be adjusted to hold hacksaw blades of different lengths say from 200mm to 300mm.



2.7 HACKSAW BLADES

Hack saw blades are made of high carbon steel, low alloy steel or high speed steel. They are then hardened and tempered. They are made as thin sheet with cutting edges present on one side or on both sides. The size of the blades is specified by the distance between the holes on either sides along the length.

According to the distance between two successive teeth on the blade (pitch), they are classified as coarse, medium and fine path blades. Soft materials like plastics are cut by coarse pitch blades. Medium pitch blades are employed to cut tool steel, hard light alloys, thick sections and tubes. Materials of small thickness are cut accurately by fine pitch blades.

Reason for the breakage of Hacksaw blades,

1. The cutting action may not be of uniform speed and thrust.
2. Improper fitting of blades (improper tightness or looseness)
3. Putting into use new blades on old cuts.

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